



Lecture #18: State Classification and Limit Behavior

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Analyzing a 2 state Markov Chain

Now let's try to use this method to a general 2 state Markov chain, where the two states are "1" and "2". Let a be the probability of going from 1 to 2, and b be the probability of going from 2 to 1. Since each row has to sum to 1, this means the general chain looks like

$$\begin{array}{cc}
 & \begin{array}{cc} \mathbf{1} & \mathbf{2} \end{array} \\
 \begin{array}{c} \mathbf{1} \\ \mathbf{2} \end{array} & \begin{array}{cc} 1-a & a \\ b & 1-b \end{array}
 \end{array}$$

Now to find the stationary distribution, we form the equations

This means we get the following equations:





$$(1 - a)\pi_1 + b\pi_2 = \pi_1$$

$$a\pi_1 + (1 - b)\pi_2 = \pi_2$$

Using substitution, we arrive at

Lastly, we check that this formula is consistent with the answer that we found for the weather chain problem.





Analyzing a 3 state Markov Chain

Now we turn our attention to 3-state Markov chains, such as the social mobility problem.

$$0.7\pi_1 + 0.3\pi_2 + 0.2\pi_3 = \pi_1$$

$$0.2\pi_1 + 0.5\pi_2 + 0.4\pi_3 = \pi_2$$

$$0.1\pi_1 + 0.2\pi_2 + 0.4\pi_3 = \pi_3$$





$$-0.3\pi_1 + 0.3\pi_2 + 0.2\pi_3 = 0$$

$$0.2\pi_1 - 0.5\pi_2 + 0.4\pi_3 = 0$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

We note that the third equation implies $\pi_3 = 1 - \pi_1 - \pi_2$ and substituting this in the first two gives

Multiplying the first equation by 0.9 and adding 0.1 times the second gives

Multiplying the first equation by 0.2 and adding -0.5 times the second gives

Since the three probabilities add up to 1,



Stationary Distribution made Easy

This might have been hard with all of the computation, so let's make a computer do the hard work. If we take the above system of equations, we can rewrite it as a matrix equation:

$$\begin{cases} -0.3\pi_1 + 0.3\pi_2 + 0.2\pi_3 = 0 \\ 0.2\pi_1 - 0.5\pi_2 + 0.4\pi_3 = 0 \\ \pi_1 + \pi_2 + \pi_3 = 1 \end{cases} \implies$$





This means to solve this problem, we need to solve this equation: Let

$$\mathbf{A} = \begin{pmatrix} -0.3 & 0.2 & 1 \\ 0.3 & -0.5 & 1 \\ 0.2 & 0.4 & 1 \end{pmatrix}.$$

Then we have $\pi\mathbf{A} = (0 \ 0 \ 1)$, so we need to multiply both sides by $\mathbf{A}^{-1}$ on the right to get

$$(\pi_1 \ \pi_2 \ \pi_3) = (0 \ 0 \ 1)\mathbf{A}^{-1}.$$

Using MATLAB, we can easily compute this to get

$$\pi = (0.468085 \ 0.340425 \ 0.191489) = \left(\frac{22}{47} \quad \frac{16}{47} \quad \frac{9}{47}\right).$$





Another 3 state Example: Laundry

Example

Suppose there are three types of laundry detergent, 1, 2, and 3, and let X_n be the brand chosen on the n^{th} purchase. Customers who try these brands are satisfied and choose the same thing again with probabilities 0.8, 0.6, and 0.4 respectively. When they change they pick one of the other two brands at random. The transition probability is

	1	2	3
1	0.8	0.1	0.1
2	0.2	0.6	0.2
3	0.3	0.3	0.4

Using the first two equations and the fact that the sum of the π_i 's is 1:



Subtracting π_1 from both sides of the first equation and p_{12} from both sides of the second, this translates into $\pi\mathbf{A} = (0 \ 0 \ 1)$, with

$$\mathbf{A} = \begin{pmatrix} -0.2 & 0.1 & 1 \\ 0.2 & -0.4 & 1 \\ 0.3 & 0.3 & 1 \end{pmatrix}$$

Note that here and in the previous example the first two columns of $\mathbf{A}$ consist of the first two columns of the transition probability with 1 subtracted from the diagonal entries, and the final column is all 1's. Computing the inverse and reading the last row gives

$$\pi = (0.545454 \quad 0.272727 \quad 0.181818) = \left(\frac{6}{11} \quad \frac{3}{11} \quad \frac{2}{11} \right).$$

To check this we note that



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Remark

If y is transient, then as you take more and more steps, the probability of returning to that state get smaller and smaller until it becomes zero. Mathematically, we have $p^n(x, y) \rightarrow 0$.

This means that we can restrict our attention to recurrent states and in view of the decomposition theorem, Theorem 1.8, to chains that consist of a single irreducible class of recurrent states. Our first example shows one problem that can prevent the convergence of $p^n(x, y)$.





Motivation

Example (Ehrenfest chain)

Let X_n be the number of balls in the “left” urn after the n^{th} draw. It should be clear that X_n has the Markov property; i.e., if we want to guess the state at time $n + 1$, then the current number of balls in the left urn X_n , is the only relevant information from the observed sequence of states $X_n, X_{n-1}, \dots, X_1, X_0$. To check this we note that:

$$\mathbb{P}(X_{n+1} = i + 1 | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \frac{N - i}{N}$$

since to increase the number we have to pick one of the $N - i$ balls in the other urn. The number can also decrease by 1 with probability i/N . In symbols, we have computed that the transition probability is given by

$$p(i, i + 1) = \frac{N - i}{N}, \quad p(i, i - 1) = \frac{i}{N} \quad \text{for } 0 \leq i \leq N.$$



When $N = 3$, we get

In the second power of p the zero pattern is shifted:

Question

Will this persist?





Periodicity

Definition

The period of a state is the largest number that will divide all the $n \geq 1$ for which $p^n(x, x) > 0$. That is, it is the greatest common divisor of $I_x = \{n \geq 1 : p^n(x, x) > 0\}$.

Remark

To check that this definition works correctly, we note that in Example 1.21, $\{n \geq 1 : p^n(x, x) > 0\} = \{2, 4, \dots\}$, so the greatest common divisor is 2.





Playing with Periodicity

Example (Triangle and Square)

	-2	-1	0	1	2	3	
-2	0	0	1	0	0	0	0
-1	1	0	0	0	0	0	0
0	0	0.5	0	0.5	0	0	0
1	0	0	0	0	0	1	0
2	0	0	0	0	0	0	1
3	0	0	1	0	0	0	0

In words, from 0 we are equally likely to go to 1 or -1. From -1 we go with probability one to -2 and then back to 0, from 1 we go to 2 then to 3 and back to 0. The name refers to the fact that $0 \rightarrow -1 \rightarrow -2 \rightarrow 0$ is a triangle and $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$ is a square



Clearly, $p^3(0, 0) > 0$ and $p^4(0, 0) > 0$ so $3, 4 \in I_0$. We need to think to realized that

While periodicity is a theoretical possibility, it rarely manifests itself in applications, except occasionally as an odd-even parity problem, e.g., the Ehrenfest chain. In most cases we will find (or design) our chain to be *aperiodic*, i.e., all states have period 1. To be able to verify this property for examples, we need to discuss some theory.





Lemma

If $p(x, x) > 0$, then x has period 1.

Lemma

If $\rho_{xy} > 0$ and $\rho_{yx} > 0$ then x and y have the same period.

Remark

The short answer is that if the two states have different periods, then by going from x to y , from y to y in the various possible ways, and then from y to x , we will get a contradiction.

Theorem (Convergence Theorem)

Let p be irreducible, aperiodic, and there is a stationary distribution π . Then as $n \rightarrow \infty$, $p^n(x, y) \rightarrow \pi(y)$.